TECHNICAL REPORT

Sri Lankan Vehicle Listing Price Prediction

NGBoost probabilistic regression and comparative model evaluation

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Abstract

This study investigates NGBoost probabilistic regression and evaluates seven regression approaches for predicting historical advertised vehicle prices in Sri Lanka. The dataset contains 9,788 listings collected between December 2024 and February 2025. Training-only exploration informs model selection, while temporal validation and feature-group purging reduce leakage. The comparison includes Random Forest, XGBoost, CatBoost, LightGBM, NGBoost, Ridge and a Random Forest-XGBoost stack with a Ridge meta-model. Random Forest achieves the lowest validation MAE of 7.15 lakhs and a test MAE of 5.83 lakhs. A deployed Streamlit application supports single-vehicle inference and interactive inspection of the results, including NGBoost partial-dependence plots and permutation importance. Because the test set was examined during earlier development, the expanded findings are exploratory and require confirmation using newly collected listings.

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1. Objective and local relevance

Predict historical advertised vehicle prices from tabular listing characteristics. Buyers and sellers can explore how vehicle characteristics relate to asking prices, while users can compare models and inspect their limitations. It does not estimate verified transaction prices or current 2026 market values.

2. Dataset and provenance

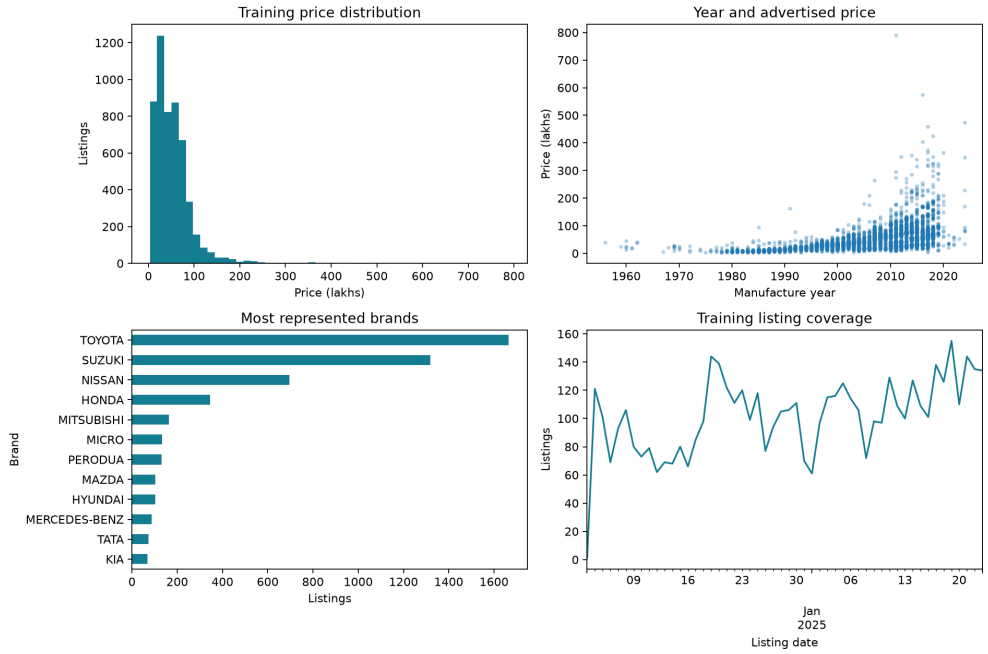
Prasad Nirmal, SriLankan Vehicle(Car) Price Dataset, version 3. Publisher metadata identifies Ikman.lk and Riyasewana.com as sources and lists MIT licensing. Original CSV and metadata are retained in data/raw. SHA-256: 4208d2673cf4d9c8d3d6042dc8a1b97004671812441975f5879550059fbce1ab. There are 9,788 source rows, spanning 2024-12-03 to 2025-02-05. Price is in LKR lakhs, confirmed by the project owner; 1 lakh is LKR 100,000.

3. Exploratory data analysis

Exploration in scripts/analyze_data.py uses only the training partition. The training price median is 42.5 lakhs, 95th percentile 120.0, maximum 790.0, and skewness 3.80. The long upper tail motivates reporting RMSE and price-band errors alongside MAE. Extreme observations remain in evaluation rather than being removed to improve scores.

Feature	Distinct values	Categories with fewer than 3 rows	Rows in rare categories
brand	48	14	18
model	1061	758	893
gear	2	0	0
fuel	4	0	0
town	106	6	6
leasing	2	0	0
condition	2	0	0
air_condition	2	0	0
power_steering	2	0	0
power_mirror	2	0	0
power_window	2	0	0

Many model labels are rare, motivating native categorical handling and regularized one-hot alternatives. Year/price Spearman association is 0.65; engine capacity/price association is 0.07. These are marginal associations, not causal effects or sufficient reasons to remove an interacting feature.



Training-only exploration

4. Data preparation and leakage controls

18 exact duplicate rows were removed after category trimming, and 0 rows failed target/date/year checks. No literal source nulls were found, but the publisher already preprocessed the table; missing original observations and prior imputation cannot be reconstructed. Mileage equals (2025 - year) * 11000 throughout the source and is excluded. The exported index, date, target and group identifiers are not predictors.

The latest record is reserved for a single-example demo, excluding its feature signature from development. Test dates start on 2025-01-24. Matching feature groups are removed from earlier training data (411 rows). There are 5,285 training, 4,073 test and one demo row. Feature signatures are a conservative proxy; without original listing IDs they cannot establish true vehicle identity. Three expanding temporal folds are used for parameter search, with matching validation groups purged from each training fold.

5. Model selection and preprocessing

- **Random Forest:** bagging reference for nonlinear interactions and variance reduction.

- **XGBoost:** regularized sequential boosting for nonlinear interactions.

- **CatBoost:** native categorical processing is relevant to rare vehicle model labels.

- **LightGBM:** an efficient histogram-based boosting alternative with bounded leaf growth.

- **NGBoost:** the principal new-method study; natural-gradient boosting of Normal distribution parameters.

- **Ridge:** additive linear comparison to test whether tree complexity adds value.

- **Stacking:** fixed RF and XGBoost pipelines combined through scaled Ridge regression.

- **Training median:** no-feature reference benchmark, not a selectable app model.

All preprocessing is fitted within folds. RF, XGBoost, LightGBM, NGBoost and Ridge use numeric median and categorical mode imputation plus one-hot encoding with rare category pooling. Ridge scales numeric inputs; tree models do not require scaling. CatBoost receives string categories with a missing marker and fold-fitted numeric median imputation. No globally fitted target encoder is used.

RF, XGBoost, CatBoost and LightGBM each compare three explicitly constrained configurations. Ridge compares alpha 1, 10 and 100. Actual settings and every search result are in experiments.csv; bounded searches are not exhaustive optimization. CatBoost's adapter supplies categorical names during fit for sklearn compatibility. LightGBM uses ordinary GBDT with row sampling, not a replication of GOSS. Stacking uses inner purged temporal out-of-fold predictions to train its Ridge meta-model (alpha 10), then refits bases on the training partition. Outer temporal folds evaluate the complete stack. Its bases are fixed, not selected with outer validation labels. Earliest rows without out-of-fold predictions do not train Ridge.

5.1 NGBoost methodology and explainability

NGBoost: probabilistic vehicle-price regression

NGBoost is the principal method studied in this extension. It uses shallow trees and natural-gradient updates to learn the mean and scale of a conditional Normal price distribution. Its training objective is negative log-likelihood; configuration selection uses validation MAE so point predictions remain comparable to the other models. Unlike Random Forest's averaging or conventional point-prediction boosting, it updates multiple distribution parameters using a geometry-adjusted gradient. This is a non-deep-learning method. See Duan et al. (2020).

Design. Fold-fitted median/mode imputation and dense one-hot encoding with rare category pooling; depth-3 base trees, minimum leaf size 10, learning rate 0.03, seed 42, and 100/200/300 rounds. Selected settings: {"model__n_estimators": 300}. Normal is an explicit simplifying assumption, not evidence that car prices are Normal.

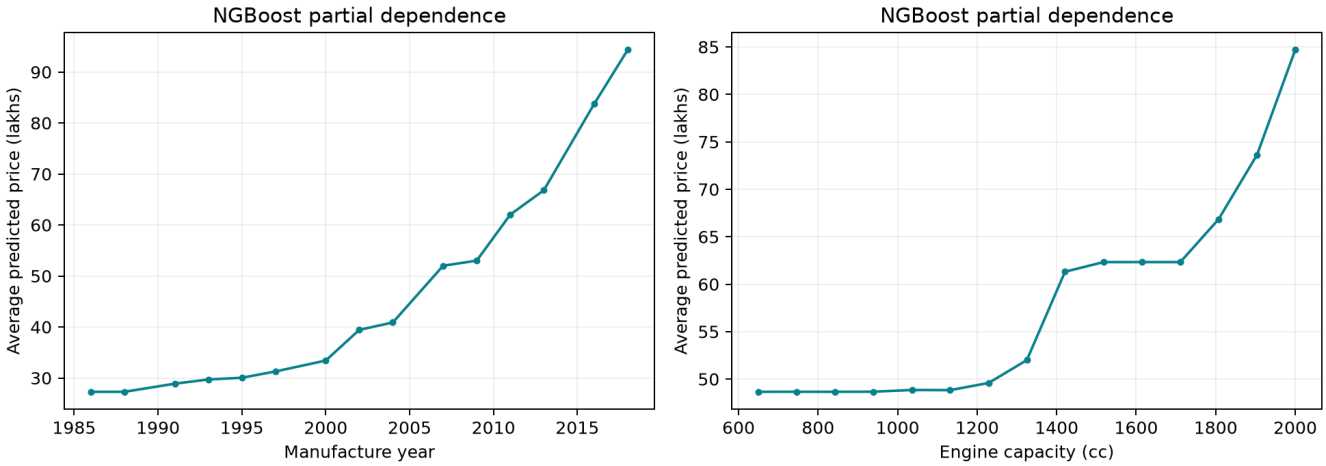
Measured performance. Validation MAE 9.43 lakhs; test MAE 8.58 lakhs; RMSE 16.45 lakhs; R2 0.861. The validation-selected recommendation is Random Forest. The research focus and the best point-prediction model need not be the same.

Global explanation. The four strongest permutation-importance fields are year, brand, engine cc, gear. Importance is the increase in test MAE after shuffling; it describes model reliance rather than causation or a direction of effect.

Partial-dependence interpretation. Manufacture year: the average model prediction changes from 27.28 to 94.33 lakhs across 1986-2018. Engine capacity (cc): the average model prediction changes from 48.66 to 84.69 lakhs across 650-2000. Newer vehicles often have higher asking prices, and engine capacity can track vehicle class. The plotted response should be compared with these expectations; non-monotonic sections may reflect model interactions, sampling and sparse data. The curves average over 256 training vehicles and cover the training 5th-95th percentiles. Varying one feature can create implausible brand/model/year/engine combinations. They are not causal depreciation curves or individual explanations.

Distribution check. Test negative log-likelihood is 3.741 (in lakh units). Nominal 90% intervals contain 89.9% of observed test prices, with mean width 33.61 lakhs. 2.4% have negative lower bounds. Coverage must be considered alongside width: a wide range can cover many prices without being useful. These intervals have not been independently calibrated and are not displayed as guaranteed price ranges in the prediction form. Negative bounds expose a limitation of the Normal assumption for positive prices.

Evidence limits. The test set was examined before this extension; results are exploratory. No test-driven retuning was performed. New, later listings are needed for independent evaluation and calibration. Better uncertainty or point accuracy than existing models is not assumed.



NGBoost partial-dependence curves

6. Evaluation and model justification

Mean validation MAE is the selection rule fixed before the expanded run. MAE expresses average monetary error directly; RMSE emphasizes large mistakes and R2 is unitless, not percentage accuracy. Fold standard deviation is descriptive, not a confidence interval. Timing includes parameter search and refitting on this machine; hardware-specific timings are not published-paper speed comparisons.

Predictive performance

Model	Validation MAE (lakhs)	Fold SD (lakhs)	Test MAE (lakhs)	Test RMSE (lakhs)	Test R2
Random Forest	7.148	0.146	5.830	14.440	0.893
Stacking	7.299	0.249	6.475	14.411	0.894
XGBoost	7.562	0.312	6.821	14.241	0.896
CatBoost	8.740	0.364	8.414	15.606	0.875
LightGBM	9.410	0.230	7.590	15.283	0.880
NGBoost	9.435	0.461	8.581	16.446	0.861
Ridge	11.124	0.671	9.994	20.694	0.781

Computational performance

Model	Training (s)	Prediction (ms/row)
Random Forest	8.168	0.011
Stacking	10.281	0.015
XGBoost	1.404	0.004
CatBoost	4.529	0.002
LightGBM	1.826	0.007
NGBoost	37.045	0.355
Ridge	0.259	0.003

Model choice and evidence

Random Forest is selected because it has the lowest mean validation MAE (**7.15 lakhs**) among the evaluated configurations. Test MAE is **5.83 lakhs**, compared with **28.67** for a training-median prediction (79.7% lower error on this historical split). This demonstrates improvement over a simple local benchmark, not over external studies.

The training data has **1,061 model labels**, including **758** occurring fewer than three times. Median training price is **42.5 lakhs** and 95th percentile **120.0 lakhs**. CatBoost tests native categorical handling; LightGBM tests another efficient nonlinear approach; Ridge provides a simpler additive reference. RF, XGBoost and stacking provide complementary ensemble comparisons.

Selection uses MAE because it measures average monetary error directly. RMSE reveals the influence of large mistakes; R2 is not percentage accuracy. Runtime and fold variation are supporting evidence, not alternate post-hoc selection rules. A more complex model is not automatically better. Error by actual-price band and unseen model label must be considered before trusting an individual estimate.

Limit: this test split was already inspected during earlier development. Expanded results are exploratory. Newly collected future listings are required for independent confirmation. The source covers December 2024-February 2025; estimates are historical asking prices rather than current valuations or verified sale prices.

7. Error analysis

The band table below uses actual target prices for retrospective analysis only; those bands are not available when making a new prediction. Counts matter: small subgroups cannot establish reliable group-level performance.

Model	Actual price band (lakhs)	Listings	MAE (lakhs)	RMSE (lakhs)
Random Forest	0-25	938	3.984	7.623
Random Forest	25-50	1307	4.025	7.912
Random Forest	50-100	1467	5.240	9.774
Random Forest	100-200	299	12.406	21.292
Random Forest	>200	62	54.080	83.984
XGBoost	0-25	938	5.191	9.362
XGBoost	25-50	1307	4.984	8.400
XGBoost	50-100	1467	6.295	10.749
XGBoost	100-200	299	14.018	21.792
XGBoost	>200	62	47.974	74.062
CatBoost	0-25	938	5.949	9.824
CatBoost	25-50	1307	5.760	8.970
CatBoost	50-100	1467	7.798	12.323
CatBoost	100-200	299	21.727	27.766
CatBoost	>200	62	52.043	74.383
LightGBM	0-25	938	5.452	9.402
LightGBM	25-50	1307	5.375	8.885
LightGBM	50-100	1467	7.081	11.588
LightGBM	100-200	299	17.041	24.746
LightGBM	>200	62	53.081	78.813
NGBoost	0-25	938	6.498	10.869
NGBoost	25-50	1307	6.652	10.492
NGBoost	50-100	1467	7.639	11.742

Model	Actual price band (lakhs)	Listings	MAE (lakhs)	RMSE (lakhs)
NGBoost	100-200	299	18.299	25.911
NGBoost	>200	62	56.183	84.616
Ridge	0-25	938	9.450	14.718
Ridge	25-50	1307	7.087	13.108
Ridge	50-100	1467	7.426	13.361
Ridge	100-200	299	19.862	26.525
Ridge	>200	62	92.647	116.693
Stacking	0-25	938	4.355	8.304
Stacking	25-50	1307	4.703	8.438
Stacking	50-100	1467	6.173	10.391
Stacking	100-200	299	13.469	22.157
Stacking	>200	62	49.293	78.588

Model	Model-label coverage	Listings	MAE (lakhs)
Random Forest	Seen model label	3484	4.830
Random Forest	Unseen model label	589	11.748
XGBoost	Seen model label	3484	5.993
XGBoost	Unseen model label	589	11.721
CatBoost	Seen model label	3484	7.664
CatBoost	Unseen model label	589	12.851
LightGBM	Seen model label	3484	6.760
LightGBM	Unseen model label	589	12.497
NGBoost	Seen model label	3484	7.779
NGBoost	Unseen model label	589	13.325
Ridge	Seen model label	3484	8.379
Ridge	Unseen model label	589	19.542
Stacking	Seen model label	3484	5.622
Stacking	Unseen model label	589	11.518

Per-brand counts and MAE for every model are saved in errors_by_brand_all_models.csv. Permutation importance measures how much test MAE changes when an input is shuffled. It explains global reliance, not causation or the reason for a single prediction. Correlated inputs can share importance; shuffling may create implausible vehicles.

8. Comparison with existing approaches

Sources checked on 14 October 2026. This is a methodological comparison, not a replication of published scores. Our controlled numerical comparisons use our own identical training/validation/test partitions.

Approach	Source evidence	Comparison with this project	Limit of the comparison
Sri Lankan used-car regression and averaging	Basnayaka (2023) compares linear regression, Random Forest, Extra Trees, gradient boosting, LightGBM and XGBoost, then averages predictions.	We evaluate related linear and tree families; retain RF/XGBoost stacking instead of simple averaging; audit synthetic mileage and use purged temporal evaluation.	The repository abstract reports "accuracy"; its metric and split are not sufficiently specified for numerical equivalence to our MAE/R2. We do not claim to reproduce or outperform it.
CatBoost	Prokhorenkova et al. (2018) present ordered boosting and categorical-feature processing.	Native categorical input is relevant to our many rare vehicle model labels; encoding is fitted within training folds.	General benchmark results do not guarantee benefits for Sri Lankan vehicle listings. Our configuration uses the library's boosting default; we do not claim to reproduce the paper's experiments.
LightGBM	Ke et al. (2017) describe efficient gradient-boosted trees, including GOSS and feature bundling.	Histogram-based LightGBM is evaluated as an efficiency-oriented nonlinear candidate, with restricted leaves and minimum leaf counts.	Our ordinary GBDT configuration uses row subsampling, not a GOSS replication. Runtime is measured locally; published speedups are not transferred to this dataset.
Median and regularized linear references	Implemented here on the same splits.	Median establishes a no-feature error level; Ridge tests whether tree complexity improves on an additive relationship.	These are local benchmarks, not reproductions of the Sri Lankan study's linear model.
NGBoost	**Duan et al. (2020) use natural gradients to learn conditional distribution parameters.**	**Normal NGBoost is the focal probabilistic regression study; point metrics, likelihood and interval coverage are evaluated on the same splits.**	**Distributional assumptions and existing test exposure limit interpretation; paper benchmark scores are not directly comparable.**

9. Single-sample inference and user interface

Reserved source row 9763: actual 30.950 lakhs, selected-model prediction 29.963, absolute error 0.987. All exported estimators are reloaded and checked against their in-memory prediction. The Streamlit interface validates fields, switches models, shows estimates in LKR, and presents model-specific importance, diagnostics, comparison and optional data exploration.

10. Deployment and repository

The application is deployed on **Streamlit Community Cloud** and uses the GitHub repository as its deployment source. The entry point is app/app.py. Saved model pipelines and their metadata support inference without retraining when a user changes the selected model.

- **Deployed application:** Sri Lankan Vehicle Price Prediction on Streamlit

- **Source repository:** Sri Lankan Vehicle Price Prediction on GitHub

Users enter vehicle characteristics, choose one of the seven trained models and request an estimated listing price in LKR. The interface also provides model comparison metrics, permutation importance, prediction diagnostics and an optional explorer for the raw and processed datasets. Model switching updates the relevant results and explanations. Estimates describe the historical listing data used in this study and should not be interpreted as verified transaction prices.

The repository contains the application, preprocessing and training code, saved model artifacts, evaluation evidence and the analysis notebook. The README provides installation and reproduction instructions. The interface specification is available in docs/interface.md.

Local verification on 28 October 2026 passed all 31 automated tests, including split isolation, all seven model pipelines, model switching, stacking and NGBoost distribution outputs. The analysis notebook executed successfully. Browser checks confirmed NGBoost inference and its model-specific explanations. These checks establish implementation functionality rather than independent predictive validity. The last recorded cloud verification before this extension was on 15 October 2026; cloud availability of the new model must be verified after deployment.

Access follows the application's and repository's sharing settings. Where either resource remains private, the reviewer requires authorized access.

11. Limitations and future evaluation

The test set was already examined during earlier development. New models are exploratory additions, not independently confirmed improvements. Resampling previously examined rows cannot provide an independent test. Future evaluation should use later listings with verified units, observed mileage, vehicle identifiers and, where available, sale prices. The model choice should be frozen before evaluation, with duplicate vehicle identities excluded across partitions. Data covers only about two months, is publisher-preprocessed, and may not reflect later import policies, exchange rates or supply changes. No causal, fairness, statistical-superiority or current-market accuracy claim is made.

Ethical use, representation and practical impact

The public source contains vehicle attributes and advertised prices. No seller names, phone numbers or contact details are used as predictors. Public availability alone does not establish consent for every downstream use; source attribution and licence information are retained, and linking records to individuals is outside scope.

The source represents online advertisers from two platforms over a short period, not the entire Sri Lankan vehicle market. Rare brands, unseen vehicle labels and high-price vehicles can have different error rates. Brand, town and equipment may reflect socioeconomic or geographic differences. Price-band, brand and category- coverage breakdowns expose some disparities but do not constitute a protected-group fairness audit: demographic attributes and verified identities are unavailable.

Underestimation may disadvantage sellers; overestimation may mislead buyers or support inflated asking prices. The interface is an exploratory decision aid, not a basis for lending, insurance or automated eligibility decisions. Human inspection, current comparable listings and verified condition remain

necessary. No model or explanation establishes causality. NGBoost uncertainty depends on the chosen distribution and can be poorly calibrated; displaying precision alone must not imply reliability. Future evaluation should include later data, wider regional and vehicle coverage, consent-aware collection and subgroup monitoring.

No deep learning, computer vision or reinforcement learning is used.

12. Conclusion

The study demonstrates an end-to-end workflow for historical vehicle listing-price regression, from data preparation and temporal evaluation to a deployed inference interface. Random Forest is the preferred model under the predefined validation-MAE criterion. NGBoost is the focal methodological contribution, evaluated honestly against existing benchmarks rather than assumed to be superior. XGBoost achieves a slightly lower test RMSE, while stacking does not improve validation MAE over Random Forest. The expanded comparison therefore supports evidence-based selection rather than assuming that additional model complexity produces better predictions.

Performance varies substantially across price bands and previously unseen vehicle model labels. Reliable evaluation on new listings, with verified vehicle identities and source measurements, is the principal next step before broader practical use.

13. References

1. Basnayaka, D. (2023). Prediction of Market Price for Used Cars in Sri Lanka Based on Machine Learning Approaches. IIT repository abstract.

2. Prokhorenkova, L., Gusev, G., Vorobev, A., Dorogush, A. V., and Gulin, A. (2018). CatBoost: unbiased boosting with categorical features. NeurIPS.

3. Ke, G. et al. (2017). LightGBM: A Highly Efficient Gradient Boosting Decision Tree. NeurIPS.

4. CatBoost categorical-feature documentation. Supports using native categorical input rather than external one-hot encoding.

5. LightGBM parameter-tuning documentation. Supports constraining leaves and minimum observations to control complexity.

6. Duan, T. et al. (2020). NGBoost: Natural Gradient Boosting for Probabilistic Prediction. ICML, PMLR 119:2690-2700.

7. NGBoost usage documentation. Distributional regression, scoring rules and prediction APIs.

Appendix A. Model-specific strengths, limitations and results

Overview

Random Forest is recommended by validation MAE. All models use the same 4,073 held-out listings. MAE measures average error; RMSE emphasizes large errors; R2 is unitless. See the comparison table for measured results, not claims of statistical superiority.

Additional model candidates and stacking are exploratory additions after the test set was examined. Stacking uses fixed base configurations and a Ridge meta-model trained on inner temporal out-of-fold predictions. Outer temporal folds evaluate the full stacking procedure. The test set is unchanged and was already examined before stacking was added; results are exploratory. All models describe historical asking prices, not completed sales or current valuations.

Random Forest: strengths and limitations

Strengths. Averages many trees to reduce prediction variance. Captures nonlinear relationships and interactions.

Limitations. Rare vehicles can have large errors. Cannot reliably extrapolate beyond training prices.

Measured results. Validation MAE: **7.15 lakhs**; test MAE: **5.83 lakhs**; test RMSE: **14.44 lakhs**; test R2: **0.893**. Training MAE was **2.68 lakhs**. Lower MAE/RMSE is better; higher R2 is better. R2 is unitless, not percentage accuracy.

Feature-importance chart. The strongest fields are **year, brand, gear, engine cc**, in that order. Longer bars mean shuffling that feature increased test error more. This is overall model reliance, not proof of causation, the direction of an effect, or an explanation for a single car. Correlated fields can share importance, and shuffling can produce unrealistic combinations.

Actual-versus-predicted graph. Points on the dashed diagonal are correct predictions. Points below it are underestimates; points above it are overestimates. Each point represents one unseen test listing.

Residual graph. Residual = actual price - predicted price. Positive values mean underestimation; negative values mean overestimation. A tight cloud around zero is desirable. The mean signed residual is **0.08 lakhs**; the largest absolute test error is **317.34 lakhs**. These describe the whole test set, not a guaranteed range for the current vehicle.

What the graphs show here. Most listings form a dense cluster at the lower end of the price range, broadly following the diagonal. The diagnostics show the direction and size of errors for this model. Points far from the diagonal or zero line identify large errors; no model is guaranteed reliable across all vehicles. The plots alone do not establish why those listings were difficult.

Large-error interpretation. RMSE (14.44) is much larger than MAE (5.83), consistent with some large errors having a strong effect on the squared-error metric. Inspect isolated points far from the diagonal/zero line; an average score can hide these failures.



Random Forest diagnostic graphs

Stacking: strengths and limitations

Strengths. Combines Random Forest and XGBoost predictions using Ridge. Meta-model learns from later out-of-fold predictions.

Limitations. Similar base-model errors can limit improvement. More training work; early rows are excluded from meta-training.

Measured results. Validation MAE: **7.30 lakhs**; test MAE: **6.47 lakhs**; test RMSE: **14.41 lakhs**; test R2: **0.894**. Training MAE was **4.78 lakhs**. Lower MAE/RMSE is better; higher R2 is better. R2 is unitless, not percentage accuracy.

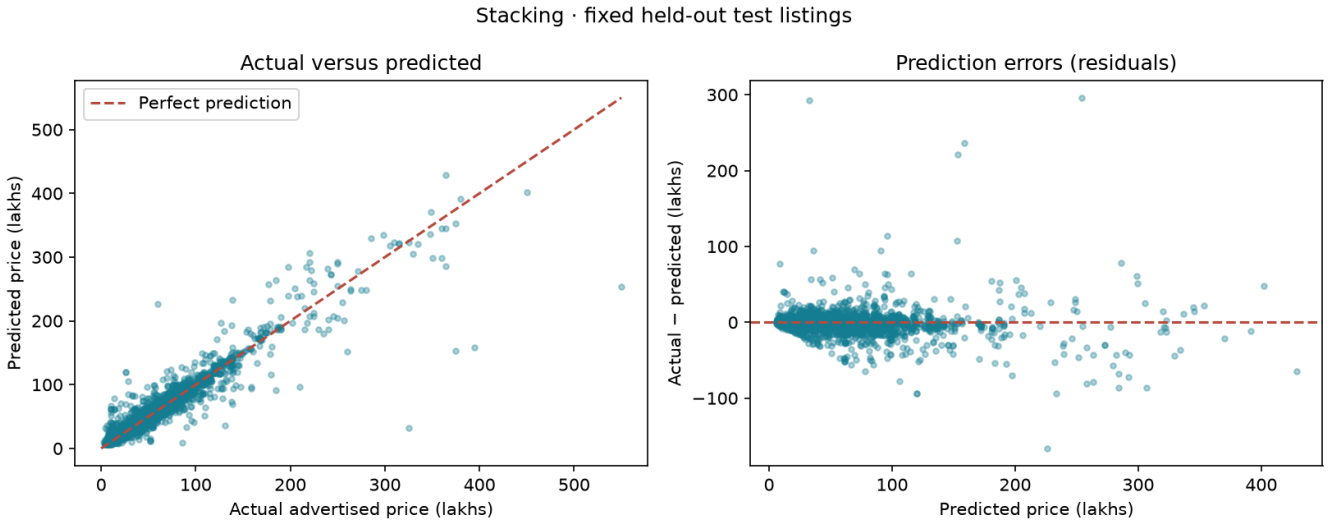
Feature-importance chart. The strongest fields are **year, brand, engine cc, gear**, in that order. Longer bars mean shuffling that feature increased test error more. This is overall model reliance, not proof of causation, the direction of an effect, or an explanation for a single car. Correlated fields can share importance, and shuffling can produce unrealistic combinations.

Actual-versus-predicted graph. Points on the dashed diagonal are correct predictions. Points below it are underestimates; points above it are overestimates. Each point represents one unseen test listing.

Residual graph. Residual = actual price - predicted price. Positive values mean underestimation; negative values mean overestimation. A tight cloud around zero is desirable. The mean signed residual is **-0.90 lakhs**; the largest absolute test error is **296.06 lakhs**. These describe the whole test set, not a guaranteed range for the current vehicle.

What the graphs show here. Most listings form a dense cluster at the lower end of the price range, broadly following the diagonal. The diagnostics show the direction and size of errors for this model. Points far from the diagonal or zero line identify large errors; no model is guaranteed reliable across all vehicles. The plots alone do not establish why those listings were difficult.

Large-error interpretation. RMSE (14.41) is much larger than MAE (6.47), consistent with some large errors having a strong effect on the squared-error metric. Inspect isolated points far from the diagonal/zero line; an average score can hide these failures.



Stacking diagnostic graphs

XGBoost: strengths and limitations

Strengths. Successive trees correct earlier errors. Regularization controls complex nonlinear patterns.

Limitations. Sensitive to parameter choices. Rare categories and market changes can still cause large errors.

Measured results. Validation MAE: **7.56 lakhs**; test MAE: **6.82 lakhs**; test RMSE: **14.24 lakhs**; test R2: **0.896**. Training MAE was **5.57 lakhs**. Lower MAE/RMSE is better; higher R2 is better. R2 is unitless, not percentage accuracy.

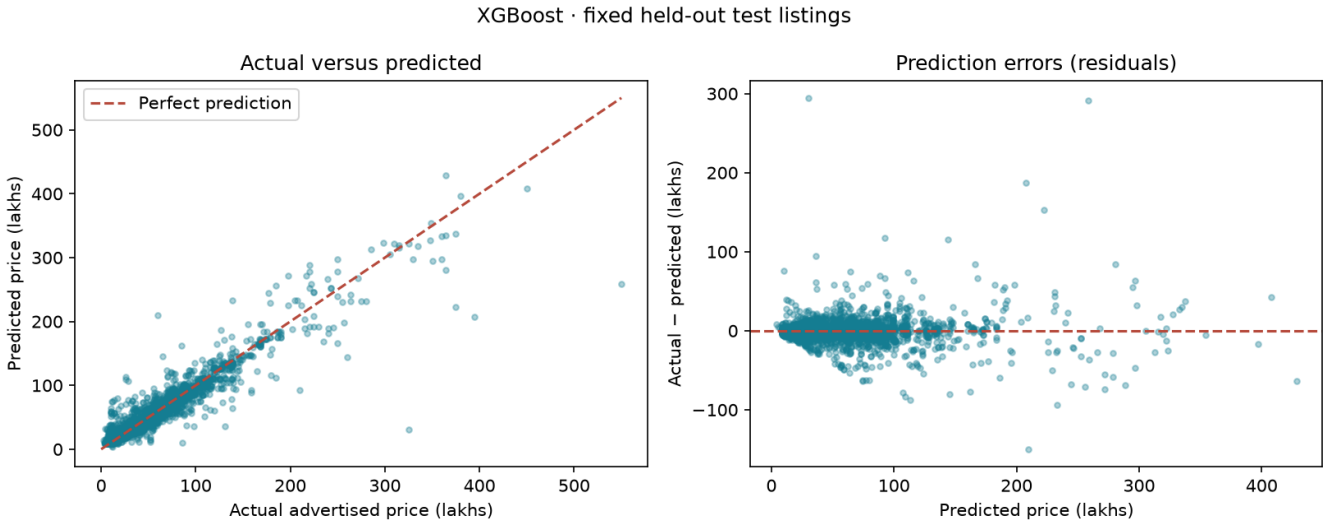
Feature-importance chart. The strongest fields are **year, brand, engine cc, gear**, in that order. Longer bars mean shuffling that feature increased test error more. This is overall model reliance, not proof of causation, the direction of an effect, or an explanation for a single car. Correlated fields can share importance, and shuffling can produce unrealistic combinations.

Actual-versus-predicted graph. Points on the dashed diagonal are correct predictions. Points below it are underestimates; points above it are overestimates. Each point represents one unseen test listing.

Residual graph. Residual = actual price - predicted price. Positive values mean underestimation; negative values mean overestimation. A tight cloud around zero is desirable. The mean signed residual is **-0.24 lakhs**; the largest absolute test error is **294.56 lakhs**. These describe the whole test set, not a guaranteed range for the current vehicle.

What the graphs show here. Most listings form a dense cluster at the lower end of the price range, broadly following the diagonal. The diagnostics show the direction and size of errors for this model. Points far from the diagonal or zero line identify large errors; no model is guaranteed reliable across all vehicles. The plots alone do not establish why those listings were difficult.

Large-error interpretation. RMSE (14.24) is much larger than MAE (6.82), consistent with some large errors having a strong effect on the squared-error metric. Inspect isolated points far from the diagonal/zero line; an average score can hide these failures.



XGBoost diagnostic graphs

CatBoost: strengths and limitations

Strengths. Uses categorical fields directly, without external one-hot encoding. Can capture patterns across many vehicle model categories.

Limitations. Rare categories still have limited supporting evidence. Categorical statistics must be learned inside training folds.

Measured results. Validation MAE: **8.74 lakhs**; test MAE: **8.41 lakhs**; test RMSE: **15.61 lakhs**; test R2: **0.875**. Training MAE was **7.76 lakhs**. Lower MAE/RMSE is better; higher R2 is better. R2 is unitless, not percentage accuracy.

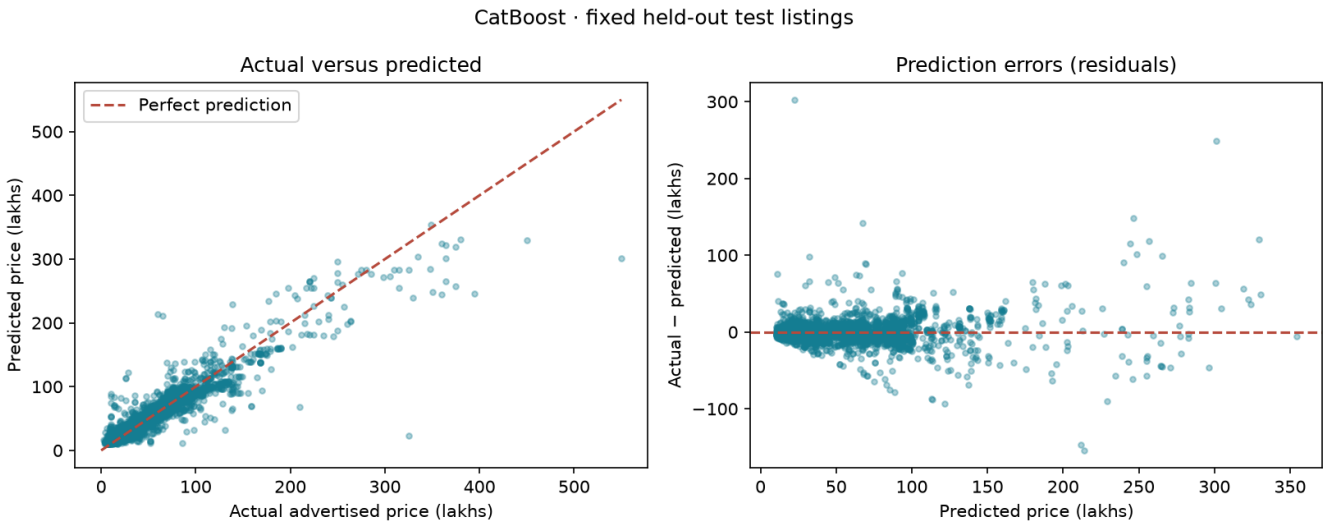
Feature-importance chart. The strongest fields are **year, brand, engine cc, gear**, in that order. Longer bars mean shuffling that feature increased test error more. This is overall model reliance, not proof of causation, the direction of an effect, or an explanation for a single car. Correlated fields can share importance, and shuffling can produce unrealistic combinations.

Actual-versus-predicted graph. Points on the dashed diagonal are correct predictions. Points below it are underestimates; points above it are overestimates. Each point represents one unseen test listing.

Residual graph. Residual = actual price - predicted price. Positive values mean underestimation; negative values mean overestimation. A tight cloud around zero is desirable. The mean signed residual is **0.11 lakhs**; the largest absolute test error is **302.58 lakhs**. These describe the whole test set, not a guaranteed range for the current vehicle.

What the graphs show here. Most listings form a dense cluster at the lower end of the price range, broadly following the diagonal. The diagnostics show the direction and size of errors for this model. Points far from the diagonal or zero line identify large errors; no model is guaranteed reliable across all vehicles. The plots alone do not establish why those listings were difficult.

Large-error interpretation. RMSE (15.61) is much larger than MAE (8.41), consistent with some large errors having a strong effect on the squared-error metric. Inspect isolated points far from the diagonal/zero line; an average score can hide these failures.



CatBoost diagnostic graphs

LightGBM: strengths and limitations

Strengths. Histogram-based boosting is an efficient tabular candidate. Leaf-wise trees capture nonlinear relationships.

Limitations. Leaf growth can overfit small groups without constraints. This implementation uses one-hot categories; unseen labels provide limited information.

Measured results. Validation MAE: **9.41 lakhs**; test MAE: **7.59 lakhs**; test RMSE: **15.28 lakhs**; test R2: **0.880**. Training MAE was **7.14 lakhs**. Lower MAE/RMSE is better; higher R2 is better. R2 is unitless, not percentage accuracy.

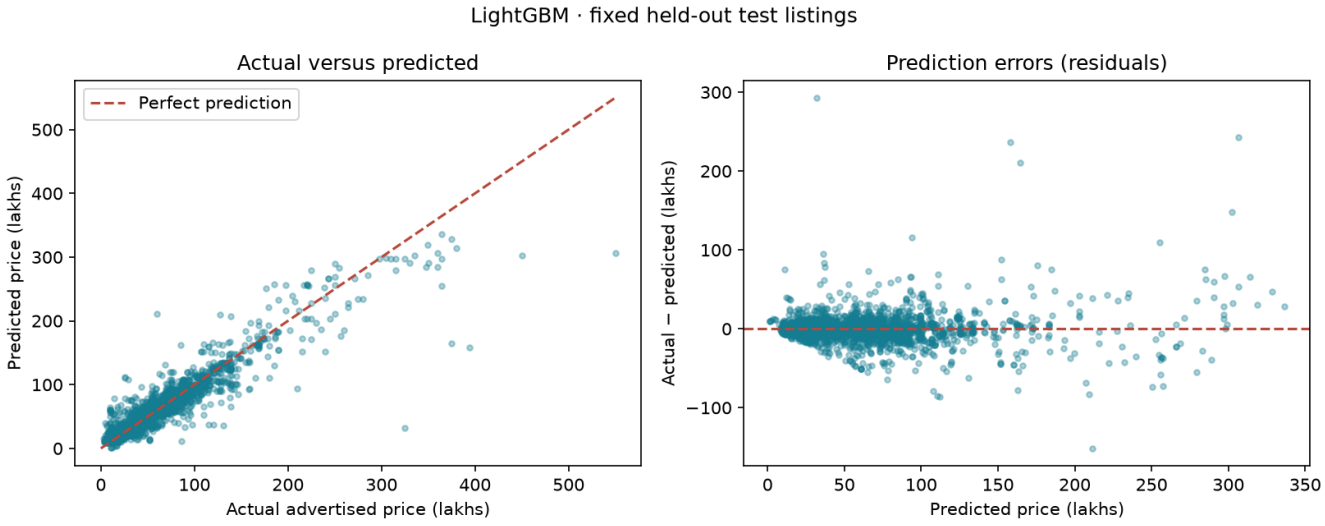
Feature-importance chart. The strongest fields are **year, brand, engine cc, gear**, in that order. Longer bars mean shuffling that feature increased test error more. This is overall model reliance, not proof of causation, the direction of an effect, or an explanation for a single car. Correlated fields can share importance, and shuffling can produce unrealistic combinations.

Actual-versus-predicted graph. Points on the dashed diagonal are correct predictions. Points below it are underestimates; points above it are overestimates. Each point represents one unseen test listing.

Residual graph. Residual = actual price - predicted price. Positive values mean underestimation; negative values mean overestimation. A tight cloud around zero is desirable. The mean signed residual is **0.02 lakhs**; the largest absolute test error is **293.00 lakhs**. These describe the whole test set, not a guaranteed range for the current vehicle.

What the graphs show here. Most listings form a dense cluster at the lower end of the price range, broadly following the diagonal. The diagnostics show the direction and size of errors for this model. Points far from the diagonal or zero line identify large errors; no model is guaranteed reliable across all vehicles. The plots alone do not establish why those listings were difficult.

Large-error interpretation. RMSE (15.28) is much larger than MAE (7.59), consistent with some large errors having a strong effect on the squared-error metric. Inspect isolated points far from the diagonal/zero line; an average score can hide these failures.



LightGBM diagnostic graphs

NGBoost: strengths and limitations

Strengths. Learns both location and scale of a price distribution using natural gradients. Provides a nonlinear probabilistic comparison to point-prediction models.

Limitations. Normal-distribution assumptions may poorly describe skewed prices and outliers. Uncertainty can be miscalibrated; negative estimates and interval bounds are possible.

Measured results. Validation MAE: **9.43 lakhs**; test MAE: **8.58 lakhs**; test RMSE: **16.45 lakhs**; test R2: **0.861**. Training MAE was **8.20 lakhs**. Lower MAE/RMSE is better; higher R2 is better. R2 is unitless, not percentage accuracy.

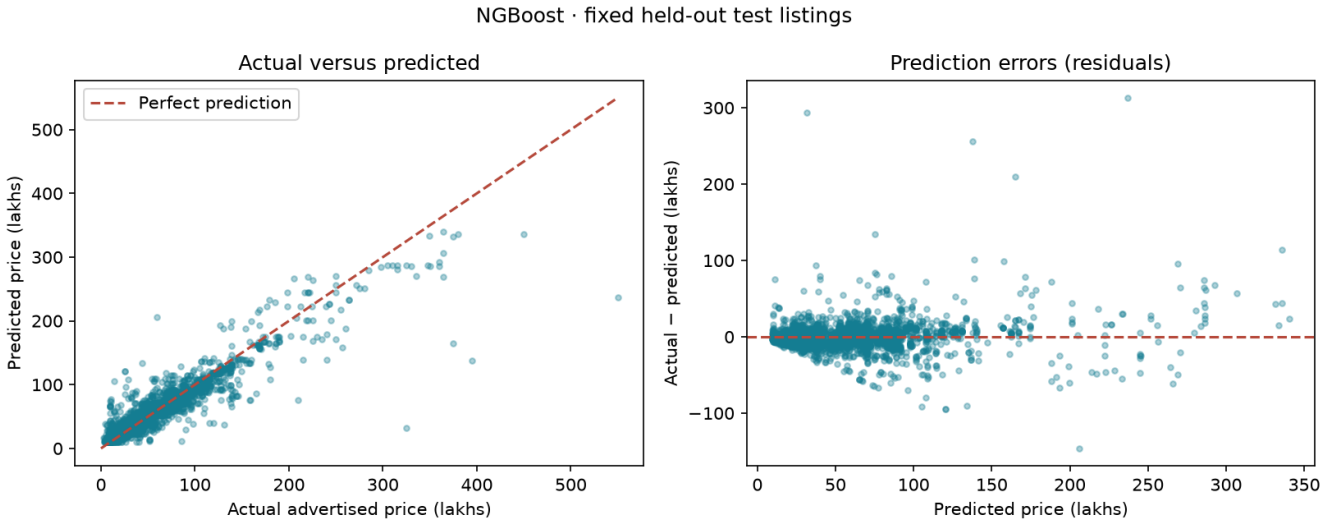
Feature-importance chart. The strongest fields are **year, brand, engine cc, gear**, in that order. Longer bars mean shuffling that feature increased test error more. This is overall model reliance, not proof of causation, the direction of an effect, or an explanation for a single car. Correlated fields can share importance, and shuffling can produce unrealistic combinations.

Actual-versus-predicted graph. Points on the dashed diagonal are correct predictions. Points below it are underestimates; points above it are overestimates. Each point represents one unseen test listing.

Residual graph. Residual = actual price - predicted price. Positive values mean underestimation; negative values mean overestimation. A tight cloud around zero is desirable. The mean signed residual is **-0.19 lakhs**; the largest absolute test error is **313.17 lakhs**. These describe the whole test set, not a guaranteed range for the current vehicle.

What the graphs show here. Most listings form a dense cluster at the lower end of the price range, broadly following the diagonal. The diagnostics show the direction and size of errors for this model. Points far from the diagonal or zero line identify large errors; no model is guaranteed reliable across all vehicles. The plots alone do not establish why those listings were difficult.

Large-error interpretation. RMSE (16.45) is much larger than MAE (8.58), consistent with some large errors having a strong effect on the squared-error metric. Inspect isolated points far from the diagonal/zero line; an average score can hide these failures.



NGBoost diagnostic graphs

Ridge: strengths and limitations

Strengths. Regularized linear reference for measuring the benefit of tree models. Simple additive structure and relatively inexpensive inference.

Limitations. Does not automatically learn nonlinear interactions. Extrapolation can produce unrealistic or negative prices.

Measured results. Validation MAE: **11.12 lakhs**; test MAE: **9.99 lakhs**; test RMSE: **20.69 lakhs**; test R2: **0.781**. Training MAE was **9.23 lakhs**. Lower MAE/RMSE is better; higher R2 is better. R2 is unitless, not percentage accuracy.

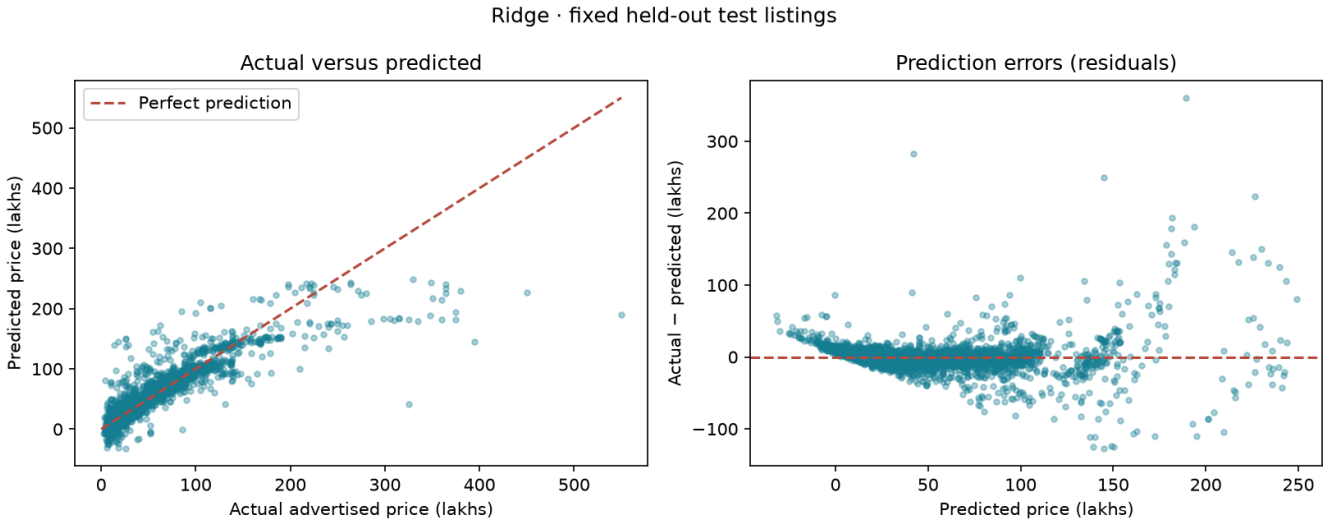
Feature-importance chart. The strongest fields are **year, brand, model, engine cc**, in that order. Longer bars mean shuffling that feature increased test error more. This is overall model reliance, not proof of causation, the direction of an effect, or an explanation for a single car. Correlated fields can share importance, and shuffling can produce unrealistic combinations.

Actual-versus-predicted graph. Points on the dashed diagonal are correct predictions. Points below it are underestimates; points above it are overestimates. Each point represents one unseen test listing.

Residual graph. Residual = actual price - predicted price. Positive values mean underestimation; negative values mean overestimation. A tight cloud around zero is desirable. The mean signed residual is **-0.39 lakhs**; the largest absolute test error is **360.61 lakhs**. These describe the whole test set, not a guaranteed range for the current vehicle.

What the graphs show here. Most listings form a dense cluster at the lower end of the price range, broadly following the diagonal. The diagnostics show the direction and size of errors for this model. Points far from the diagonal or zero line identify large errors; no model is guaranteed reliable across all vehicles. The plots alone do not establish why those listings were difficult.

Large-error interpretation. RMSE (20.69) is much larger than MAE (9.99), consistent with some large errors having a strong effect on the squared-error metric. Inspect isolated points far from the diagonal/zero line; an average score can hide these failures.



Ridge diagnostic graphs